

MAT41-4

Differential Equations

First- and second-order ODEs, numerical solution, and continuous dynamics

Modelling change over time as an equation in derivatives — solving it by hand when we can, by computer when we must.

Albert School — 26 hours

Preface

Almost every quantitative model that evolves in time is, at heart, a statement about a **rate of change**. A bank balance grows at a rate proportional to itself; a population grows quickly when small and saturates when it nears the limit of its resources; a loan shrinks as repayments outpace accruing interest. In each case we do not know the quantity directly — we know the rule that links the quantity to its own derivative. An equation of that form is an **ordinary differential equation** (ODE), and solving it means recovering the quantity from the rule.

Everything you need to read this course you already met in two earlier modules. From MAT11-1 (Foundations) come the tools of single-variable calculus we will use on every page: the derivative as a rate of change, the antiderivative and the Fundamental Theorem of Calculus that lets us undo differentiation, integration by parts and substitution, the exponential and logarithm and their derivatives, and — for second-order equations — complex numbers in the form $e^{i\theta} = \cos \theta + i \sin \theta$. From MAT21-3 (Multivariable Analysis) comes fluency with functions of several variables and with iterative numerical schemes like gradient descent; an ODE's right-hand side $f(t, y)$ is a function of **two** variables, and the numerical methods we build here advance a solution step by step in exactly the iterative spirit you already know.

The course has two complementary halves. The first is **analytical**: a small catalogue of ODE types — separable, first-order linear, second-order linear with constant coefficients — each with a method that produces an exact formula. The second is **numerical**: when no formula exists (which is the common case), we approximate the solution with Euler's method and Runge–Kutta, and learn to quantify and control the error. A recurring theme ties the two together: given a new equation, the first skill is to **judge which approach is realistic** before committing to it.

We close each idea where it pays off — in finance and growth models: continuous compounding, portfolios with contributions, logistic growth, debt repayment. This course is also the designated mathematical prerequisite for stochastic calculus, where these deterministic dynamics acquire a random term; the firmer your grasp here, the smoother that transition.

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1 What is a differential equation?

A quantity $y(t)$ changes over time. Suppose we are told that **at every instant its rate of change equals twice its current value**. In symbols, $y'(t) = 2y(t)$. This is not an equation for a number; it is an equation for an unknown **function**. A solution is any function that satisfies it for all t — here, as you can check by differentiating, $y(t) = Ce^{2t}$ for any constant C .

Definition 1 A **first-order ordinary differential equation (ODE)** in **standard form** is an equation

$$y'(t) = f(t, y(t)),$$

where f is a given function of two variables. A **solution** on an interval I is a differentiable function $y : I \rightarrow \mathbb{R}$ such that $y'(t) = f(t, y(t))$ for every $t \in I$. The equation is **ordinary** because the unknown depends on a single variable t (as opposed to a **partial** differential equation).

The word “first-order” refers to the highest derivative that appears — here just y' . Later we treat **second-order** equations, in which y'' appears.

Worked example — Reading an equation into standard form.

- $y' = 3y$ is already standard, with $f(t, y) = 3y$ (no explicit t).
- $y' + 2ty = t$ rearranges to $y' = t - 2ty$, so $f(t, y) = t - 2ty$.
- $ty' = y + t^2$ becomes $y' = \frac{y}{t} + t$, valid where $t \neq 0$; here $f(t, y) = y/t + t$.

The standard form always isolates y' on the left, exposing the function f that drives the dynamics.

A first-order ODE alone has infinitely many solutions — the constant C above is free. To single one out we must pin the value at one instant.

Definition 2 A **Cauchy problem** (or **initial value problem**) couples an ODE with an **initial condition**:

$$\begin{cases} y'(t) = f(t, y(t)) \\ y(t_0) = y_0 \end{cases}.$$

We seek the solution whose graph passes through the prescribed point (t_0, y_0) .

Worked example — From general to particular. The general solution of $y' = 2y$ is $y(t) = Ce^{2t}$. Impose $y(0) = 5$: then $Ce^0 = C = 5$, so the Cauchy problem $y' = 2y$, $y(0) = 5$ has the single solution $y(t) = 5e^{2t}$. The initial condition converts a family of curves into one curve.

1.1 The geometry: slope fields

Standard form has a vivid picture. At each point (t, y) of the plane the number $f(t, y)$ is a **slope** — the slope any solution passing through that point must have. Drawing a short segment of that slope at a grid of points produces a **slope field**, and a solution is a curve that stays tangent to the field everywhere. An initial condition drops a pin at one point; the solution threads through the field from there.

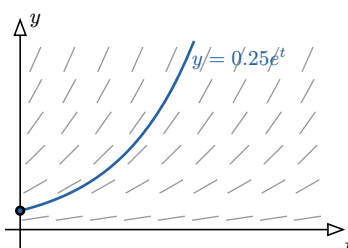


Figure 1: Slope field of $y' = y$: at each point the segment has slope equal to the height y . The blue curve is the solution with $y(0) = 0.25$; it stays tangent to the field at every point. Different initial heights give different curves, all sharing the same field.

2 Existence and uniqueness: the Cauchy–Lipschitz theorem

Before solving an equation it is worth asking two prior questions. **Does** a solution to the Cauchy problem exist? And if so, is it **unique** — or could two different curves pass through the same starting point? For modelling this is not pedantry: uniqueness is what makes a deterministic model deterministic. If the present state (t_0, y_0) failed to determine the future, the model would predict nothing.

The answer hinges on how **gently** f depends on y . A mild, quantitative form of “gentle” suffices.

Definition 3 A function $f(t, y)$ is **Lipschitz in y** on a region if there is a constant L (a **Lipschitz constant**) such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$$

for all $(t, y_1), (t, y_2)$ in the region. Intuitively: changing y changes f by no more than L times as much — f has a bounded “slope in the y direction”.

Theorem 4 (Cauchy–Lipschitz / Picard–Lindelöf) Suppose f is continuous on a rectangle around (t_0, y_0) and Lipschitz in y there. Then the Cauchy problem

$$y' = f(t, y), \quad y(t_0) = y_0$$

has **exactly one** solution on some open interval containing t_0 .

In practice you rarely produce a Lipschitz constant by hand. A continuous partial derivative does the job for you.

Proposition 5 (A practical sufficient condition) If f and its partial derivative $\partial f/\partial y$ are continuous on a region (i.e. f is C^1 in y), then f is Lipschitz in y on any closed bounded sub-rectangle, and existence–uniqueness holds there. So: **if $\partial f/\partial y$ exists and is continuous near the initial point, the Cauchy problem has a unique solution.**

Worked example — Checking the conditions. For $y' = 2y$, $f(t, y) = 2y$ and $\partial f/\partial y = 2$, continuous everywhere — unique solution through every initial point, as we found. For $y' = t - 2ty$, $\partial f/\partial y = -2t$ is continuous everywhere, so again every Cauchy problem is uniquely solvable. The check is usually this quick.

2.1 When uniqueness fails: $y' = \sqrt{|y|}$

The hypothesis is not decorative. Where $\partial f/\partial y$ blows up, uniqueness can genuinely break — and the standard cautionary tale is $y' = \sqrt{|y|}$ started at the origin.

Worked example — Two solutions from one start. Consider $y' = \sqrt{|y|}$ with $y(0) = 0$. Here $f(t, y) = \sqrt{|y|}$.

- The constant $y(t) = 0$ is plainly a solution: both sides are 0.
- But $y(t) = \frac{1}{4}t^2$ for $t \geq 0$ also works: then $y' = \frac{1}{2}t$ and $\sqrt{|y|} = \sqrt{t^2/4} = t/2$ for $t \geq 0$. Same.

Both pass through $(0, 0)$, so the Cauchy problem has (at least) **two** solutions. We can even glue them: stay at 0 until any time $a \geq 0$, then peel off along $\frac{1}{4}(t - a)^2$. There are infinitely many solutions.

What went wrong? At $y = 0$, $\partial f/\partial y = 1/(2\sqrt{|y|})$ is undefined — f has a **vertical** tangent in the y -direction, infinitely steep, so no finite Lipschitz constant exists near the origin. The hypothesis of Cauchy–Lipschitz fails exactly where uniqueness does.

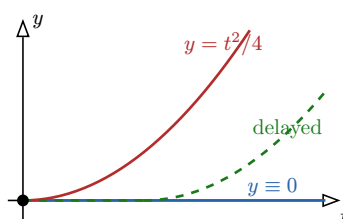


Figure 2: Non-uniqueness for $y' = \sqrt{|y|}$, $y(0) = 0$. The flat solution $y \equiv 0$ (blue), the parabola $y = t^2/4$ (red), and a “delayed” solution (green) that waits at 0 before peeling off all start at the origin. Because $\partial f/\partial y$ is infinite at $y = 0$, Cauchy–Lipschitz does not apply and uniqueness is lost.

Common pitfall Existence is the easy half: mere continuity of f guarantees **a** solution exists (Peano’s theorem). It is **uniqueness** that needs the Lipschitz condition. When you check conditions, the question to ask is always “is $\partial f/\partial y$ continuous here?” — and to be suspicious wherever f involves a square root, an absolute value, or a fractional power that can hit zero.

3 Separable equations

We now begin the analytical catalogue. The first and most directly solvable type is the one in which the two variables can be **separated** onto opposite sides of the equation.

Worked example — The idea on a concrete case. Solve $y' = 2ty$. Treat $y' = (dy)/(dt)$ as a ratio and shuffle every y to the left, every t to the right:

$$\frac{dy}{y} = 2t dt.$$

Integrate both sides: $\ln|y| = t^2 + C$. Exponentiate: $|y| = e^{t^2+C} = e^C e^{t^2}$, so $y = Ae^{t^2}$ where $A = \pm e^C$ absorbs the constant (and $A = 0$ recovers the solution $y \equiv 0$). The whole method is in this manipulation.

Definition 6 A first-order ODE is **separable** if its right-hand side factors as a function of t times a function of y :

$$y' = g(t) h(y).$$

Proposition 7 (Separation of variables) For $y' = g(t)h(y)$ with $h(y) \neq 0$, divide by $h(y)$ and integrate:

$$\int \frac{dy}{h(y)} = \int g(t) dt.$$

Evaluating both antiderivatives and solving for y gives the general solution. An initial condition $y(t_0) = y_0$ fixes the constant of integration.

Worked example — A Cauchy problem, start to finish. Solve $y' = t/y$, $y(0) = 2$. Separate: $y dy = t dt$. Integrate: $\frac{1}{2}y^2 = \frac{1}{2}t^2 + C$, i.e. $y^2 = t^2 + C'$. Apply $y(0) = 2$: $4 = 0 + C'$, so $C' = 4$ and $y^2 = t^2 + 4$. Choosing the branch with $y(0) = 2 > 0$ gives

$$y(t) = \sqrt{t^2 + 4}.$$

3.1 Equilibrium solutions

Dividing by $h(y)$ is illegal exactly where $h(y) = 0$ — and those values are not a nuisance but some of the most important solutions.

Definition 8 An **equilibrium** (or **constant**) **solution** of $y' = g(t)h(y)$ is a constant $y(t) \equiv y^*$ where $h(y^*) = 0$. Then $y' = 0$ for all t : the system, started exactly at y^* , stays there forever.

Worked example — Equilibria of logistic-type growth. For $y' = y(1 - y)$ we have $h(y) = y(1 - y)$, which vanishes at $y = 0$ and $y = 1$. So $y \equiv 0$ and $y \equiv 1$ are equilibrium solutions: a population at 0 stays extinct, one exactly at carrying capacity 1 stays put. Every other solution is found by separating, and — as we will see — slides between these two levels.

Common pitfall When you separate by dividing by $h(y)$, you silently **assume** $h(y) \neq 0$ and therefore **lose the equilibrium solutions**. Always find the roots of h first, record those constant solutions separately, and only then divide. Forgetting them is the single most common error in this method.

3.2 Domain of validity

A separable solution is rarely valid for all t . The formula may blow up, leave the region where the algebra was legal, or cross an equilibrium it can never actually reach.

Worked example — A solution that escapes to infinity. Solve $y' = y^2$, $y(0) = 1$. Separate: $-1/y = t + C$. With $y(0) = 1$, $C = -1$, so $y(t) = 1/(1 - t)$. This is a perfectly good solution — but only on the interval $t < 1$. As $t \rightarrow 1^-$ the solution escapes to $+\infty$ in **finite time**; beyond $t = 1$ it does not exist. The **domain of validity** is $(-\infty, 1)$, the largest interval containing $t_0 = 0$ on which the formula stays finite and differentiable.

Remark The domain of validity is the largest **open interval containing t_0** on which the solution is defined and stays in the region where f satisfies the Cauchy–Lipschitz hypotheses. It can be smaller than where the final formula happens to make sense: in the example $1/(1 - t)$ is also defined for $t > 1$, but that branch is a **different** solution, not the continuation of ours. Always report the interval, not just the formula.

4 First-order linear equations

The next type in the catalogue is **linear**: y and y' appear only to the first power and are not multiplied together. These are not generally separable, but a clever multiplier turns the left-hand side into a single derivative.

Definition 9 A **first-order linear ODE** is one of the form

$$y' + a(t)y = b(t),$$

with given coefficient $a(t)$ and right-hand side $b(t)$. It is **homogeneous** if $b(t) \equiv 0$ and **non-homogeneous** otherwise.

4.1 The integrating factor

The trick is to multiply through by a cleverly chosen function $\mu(t)$ so the left side becomes the derivative of a product.

Worked example — Spotting the product rule. Take $y' + y = 3$. Multiply by $\mu = e^t$:

$$e^t y' + e^t y = 3e^t.$$

The left side is exactly $(e^t y)'$ by the product rule. So $(e^t y)' = 3e^t$; integrate to get $e^t y = 3e^t + C$, hence $y = 3 + Ce^{-t}$. The factor e^t did all the work.

What made $\mu = e^t$ work is that $\mu' = \mu \cdot a$ with $a = 1$. In general we want μ with $\mu' = a(t)\mu$, which is itself a separable equation solved by an exponential.

Proposition 10 (Integrating factor method) For $y' + a(t)y = b(t)$, let the **integrating factor** be

$$\mu(t) = e^{\int a(t) dt}.$$

Then $(\mu y)' = \mu y' + \mu a y = \mu b$, so multiplying through and integrating,

$$\mu(t)y(t) = \int \mu(t)b(t) dt + C, \quad \text{hence} \quad y(t) = \frac{1}{\mu(t)} \left(\int \mu(t)b(t) dt + C \right).$$

Worked example — A non-constant coefficient. Solve $y' + 2ty = t$. Here $a(t) = 2t$, so $\mu = e^{\int 2t dt} = e^{t^2}$. Multiply: $(e^{t^2}y)' = te^{t^2}$. Integrate the right side by substitution ($u = t^2$): $\int te^{t^2} dt = \frac{1}{2}e^{t^2} + C$. So $e^{t^2}y = \frac{1}{2}e^{t^2} + C$ and

$$y(t) = \frac{1}{2} + Ce^{-t^2}.$$

Common pitfall When you compute $\mu = e^{\int a dt}$, take **any one** antiderivative of a — drop the “ $+C$ ” here. (A different choice multiplies μ by a constant, which cancels out and changes nothing.) The single arbitrary constant of the problem belongs to the **final** integration, not to the integrating factor.

4.2 Variation of the constant

A second, equivalent route splits the problem in two and explains **why** the general solution looks the way it does. It generalises to systems and to higher order, so it is worth seeing.

First solve the **homogeneous** equation $y' + a(t)y = 0$ by separation: $y_{h(t)} = Ce^{-\int a dt} = C/\mu(t)$. Now “vary the constant”: look for a particular solution of the full equation in the form $y_{p(t)} = C(t)/\mu(t)$, promoting the constant C to a function $C(t)$.

Proposition 11 (Variation of the constant) Substituting $y = C(t)/\mu(t)$ into $y' + ay = b$ makes the a -terms cancel and leaves $C'(t) = \mu(t)b(t)$, so $C(t) = \int \mu b dt$. The general solution is then

$$y(t) = \underbrace{\frac{C_0}{\mu(t)}}_{\text{homogeneous}} + \underbrace{\frac{\int \mu b dt}{\mu(t)}}_{\text{particular}},$$

the same answer the integrating factor gives, now displayed as **general homogeneous + one particular**.

The structure **general solution = homogeneous solutions + one particular solution** is the central fact about all linear equations, first or second order. The homogeneous part carries the free constants (the “memory” of initial conditions); the particular part answers the forcing $b(t)$. Keep the two roles distinct and linear ODEs become bookkeeping.

5 Second-order linear equations with constant coefficients

Many physical and financial models involve acceleration, not just velocity — a second derivative. We treat the most important solvable family: linear, with **constant** coefficients.

Definition 12 A second-order linear ODE with constant coefficients is

$$ay'' + by' + cy = g(t), \quad a \neq 0,$$

with real constants a, b, c . It is **homogeneous** when $g(t) \equiv 0$. A Cauchy problem now needs **two** initial conditions, typically $y(t_0)$ and $y'(t_0)$, because there are two free constants.

5.1 The homogeneous equation and the characteristic equation

Solve $ay'' + by' + cy = 0$. Guided by the first-order case, where solutions were exponentials, try $y = e^{rt}$. Then $y' = re^{rt}$, $y'' = r^2e^{rt}$, and substituting,

$$(ar^2 + br + c)e^{rt} = 0.$$

Since $e^{rt} \neq 0$, the exponent r must be a root of a quadratic.

Definition 13 The **characteristic equation** of $ay'' + by' + cy = 0$ is

$$ar^2 + br + c = 0.$$

Its discriminant is $\Delta = b^2 - 4ac$. The form of the general solution is dictated by the sign of Δ .

The three cases — two real roots, one repeated root, a complex-conjugate pair — are exactly the three cases of the quadratic formula, and they produce three qualitatively different behaviours.

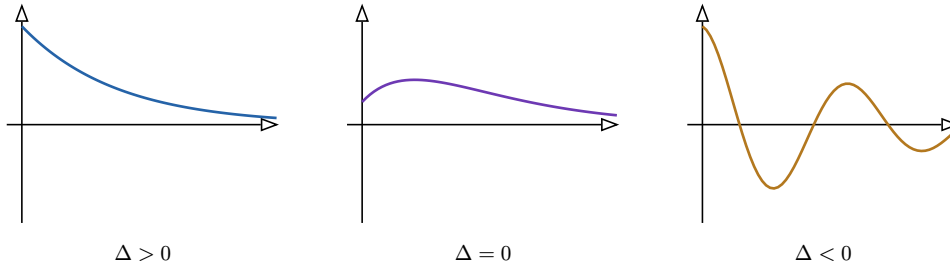


Figure 3: The three discriminant cases of $ay'' + by' + cy = 0$ (drawn for negative real parts, the damped regime). $\Delta > 0$: a sum of two real exponentials, no oscillation. $\Delta = 0$: the repeated-root form $(A + Bt)e^{rt}$. $\Delta < 0$: complex roots give a sinusoid inside an exponential envelope — damped oscillation.

Proposition 14 (Distinct real roots (\$\Delta > 0\$)) If $ar^2 + br + c = 0$ has two distinct real roots $r_1 \neq r_2$, the general solution is

$$y(t) = Ae^{r_1 t} + Be^{r_2 t}, \quad A, B \in \mathbb{R}.$$

Worked example. $y'' - 3y' + 2y = 0$ has characteristic equation $r^2 - 3r + 2 = 0$, roots $r = 1, 2$. General solution $y = Ae^t + Be^{2t}$. With $y(0) = 0$, $y'(0) = 1$: $A + B = 0$ and $A + 2B = 1$, giving $B = 1$, $A = -1$, so $y = e^{2t} - e^t$.

Proposition 15 (Repeated real root (\$\Delta = 0\$)) If the quadratic has a single repeated root $r = -b/(2a)$, the second independent solution gains a factor t :

$$y(t) = (A + Bt)e^{rt}.$$

Worked example. $y'' - 4y' + 4y = 0$: characteristic equation $r^2 - 4r + 4 = (r - 2)^2 = 0$, double root $r = 2$. General solution $y = (A + Bt)e^{2t}$.

Proposition 16 (Complex conjugate roots ($\Delta < 0$)) If the roots are complex, $r = \alpha \pm i\beta$ with $\beta = \sqrt{4ac - b^2}/(2a) > 0$, then using Euler's formula the real general solution is

$$y(t) = e^{\alpha t}(A \cos(\beta t) + B \sin(\beta t)).$$

The real part α sets growth ($\alpha > 0$) or decay ($\alpha < 0$); the imaginary part β sets the oscillation frequency.

Worked example. $y'' + y = 0$: roots $r = \pm i$ ($\alpha = 0$, $\beta = 1$). General solution $y = A \cos t + B \sin t$ — undamped oscillation, the equation of simple harmonic motion. For $y'' + 2y' + 5y = 0$: $r = -1 \pm 2i$, so $y = e^{-t}(A \cos 2t + B \sin 2t)$, a decaying oscillation.

Remark Where does the complex form become real? The two complex solutions are $e^{(\alpha+i\beta)t}$ and $e^{(\alpha-i\beta)t}$. By Euler's formula $e^{i\beta t} = \cos \beta t + i \sin \beta t$ (from MAT11-1), real linear combinations of the two collapse to $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$. Complex numbers are the bridge, but the final solution is real.

5.2 Non-homogeneous equations: undetermined coefficients

When $g(t) \neq 0$, the structure is again **general homogeneous + one particular**. We already produce the homogeneous part from the characteristic equation; we need one particular solution y_p of $ay'' + by' + cy = g(t)$. When g is a polynomial, exponential, sine/cosine, or product of these, we can **guess** y_p of the same form with undetermined coefficients and solve for them.

Proposition 17 (Method of undetermined coefficients) Look for a particular solution y_p shaped like the forcing $g(t)$:

forcing $g(t)$	trial y_p
polynomial of degree n	general degree- n polynomial
e^{kt}	Ce^{kt}
$\cos(\omega t)$ or $\sin(\omega t)$	$C \cos(\omega t) + D \sin(\omega t)$

Substitute into the equation and match coefficients to find the constants.

Worked example — A polynomial forcing. Solve $y'' - 3y' + 2y = 4t$. Homogeneous part (from before): $y_h = Ae^t + Be^{2t}$. For y_p try a linear polynomial $y_p = at + b$. Then $y_p' = a$, $y_p'' = 0$, and $0 - 3a + 2(at + b) = 2at + (2b - 3a) = 4t$. Match: $2a = 4$ so $a = 2$; $2b - 3a = 0$ so $b = 3$. Thus $y_p = 2t + 3$ and

$$y(t) = Ae^t + Be^{2t} + 2t + 3.$$

Common pitfall Resonance. If your trial y_p is already a solution of the homogeneous equation, the substitution collapses to $0 = g$ and the method fails. Cure: multiply the trial

by t (or t^2 for a double root) until it is no longer homogeneous. For $y'' - y = e^t$, the natural guess Ce^t solves the homogeneous equation, so use $y_p = Cte^t$ instead.

6 Numerical methods

Most ODEs have **no** closed-form solution — the integrals are intractable or the equation is nonlinear in a way no trick unlocks. The remedy is to abandon formulas and instead **march the solution forward** in small steps, computing approximate values $y_0, y_1, y_2, \dots$ at times $t_0, t_1, t_2, \dots$. This is the same step-by-step spirit as the iterative methods of MAT21-3.

6.1 Euler's explicit method

The simplest scheme reads the slope field literally. Standing at (t_n, y_n) , the ODE tells us the slope there is $f(t_n, y_n)$. Follow that straight line for a small **step** h and read off the new height.

Definition 18 Euler's explicit method with step size h advances by

$$t_{n+1} = t_n + h, \quad y_{n+1} = y_n + h f(t_n, y_n),$$

starting from the initial condition (t_0, y_0) . Each step is one tangent-line extrapolation.

Worked example — Three steps by hand. Apply Euler to $y' = y$, $y(0) = 1$ with $h = 0.5$ (exact solution e^t).

- $y_1 = y_0 + hy_0 = 1 + 0.5(1) = 1.5$ (exact $e^{0.5} \approx 1.649$)
- $y_2 = 1.5 + 0.5(1.5) = 2.25$ (exact $e^1 \approx 2.718$)
- $y_3 = 2.25 + 0.5(2.25) = 3.375$ (exact $e^{1.5} \approx 4.482$)

The estimate trails the true value — each tangent step undershoots a curve that bends upward — and the gap **accumulates**. Halving h would roughly halve the error, as we quantify below.

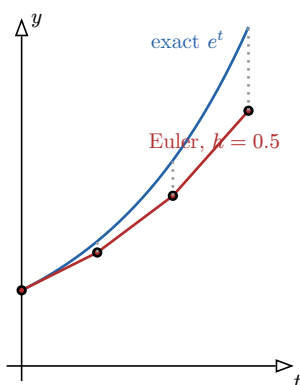


Figure 4: Euler's method on $y' = y$, $y(0) = 1$ with $h = 0.5$ (the t -axis is scaled). Each red segment is a tangent-line step; because the true curve (blue) bends upward, every step lands below it and the dotted gaps grow. Smaller h hugs the curve more closely.

6.2 Error analysis and convergence order

Two distinct errors live in a stepping scheme. The **local** error is the mistake made in a **single** step, assuming we started that step exactly right. The **global** error is the total mistake at a fixed final time after all steps accumulate.

Proposition 19 (Euler is first-order accurate) For Euler’s method, a Taylor expansion gives local error $y(t_{n+1}) - (y(t_n) + hf(t_n, y(t_n))) = \frac{1}{2}y''(\xi)h^2 = O(h^2)$ per step. Over a fixed interval the number of steps is proportional to $1/h$, so the errors compound to a **global** error

$$|y(t_{\text{final}}) - y_N| = O(h).$$

Euler is therefore a **first-order** method: halving the step roughly halves the global error.

The exponent of h in the global error is the **order** of the method. Order 1 is slow: to gain one decimal digit of accuracy you need ten times as many steps. The remedy is a higher-order scheme.

6.3 Runge–Kutta (RK4)

Euler samples the slope only at the **start** of each step. Runge–Kutta methods sample at several points within the step and combine them, cancelling more terms of the Taylor error. The classical fourth-order scheme is the workhorse.

Definition 20 The **classical Runge–Kutta method** (RK4) advances by a weighted average of four slope samples:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right), \\ k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_2\right), \\ k_4 &= f(t_n + h, y_n + h k_3), \\ y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

k_1 is the slope at the start (Euler’s), k_2, k_3 at the midpoint, k_4 at the end; the $1/6, 2/6, 2/6, 1/6$ weights are tuned to kill error terms through h^4 .

Proposition 21 (RK4 is fourth-order accurate) RK4 has local error $O(h^5)$ and **global** error $O(h^4)$. Halving the step divides the error by about $2^4 = 16$. For the same accuracy RK4 needs dramatically fewer steps than Euler — at the cost of four function evaluations per step instead of one, a trade that almost always favours RK4.

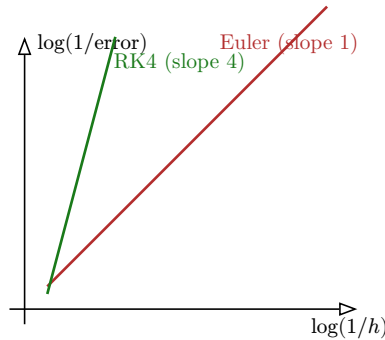


Figure 5: Empirical convergence, plotted as $\log(1/\text{error})$ against $\log(1/h)$: the slope of each line is the method's order. Euler climbs with slope 1, RK4 with slope 4 — for each tenfold decrease in h , Euler's error drops by 10, RK4's by 10^4 . This log–log slope is how order is verified empirically.

6.4 Choosing the step size

Smaller h means more accuracy but more work, and — past a point — more accumulated round-off. The practical rule is to shrink h until the answer stops changing meaningfully.

Step-doubling check. Compute the solution with step h and again with $h/2$. If the two final values agree to the precision you need, h is small enough; if not, halve again. For a p -th order method the error of the $h/2$ run is about 2^p times smaller, so the difference between the two runs is itself a usable error estimate — the basis of **adaptive** step-size control, which good library solvers do automatically.

7 Solving ODEs in Python

By hand, stepping methods are tedious and error-prone; their natural home is code. We use NumPy for arrays and SciPy's `solve_ivp` as the production solver.

7.1 Euler and RK4 by hand

Euler's method is a five-line loop — the clearest way to internalise it.

```
import numpy as np

def euler(f, t0, y0, h, n):
    t = np.empty(n + 1); y = np.empty(n + 1)
    t[0], y[0] = t0, y0
    for k in range(n):
        y[k + 1] = y[k] + h * f(t[k], y[k])    # tangent-line step
        t[k + 1] = t[k] + h
    return t, y

# y' = y, y(0) = 1 on [0, 1]; exact solution is e^t
t, y = euler(lambda t, y: y, 0.0, 1.0, h=0.1, n=10)
print(y[-1], np.exp(1))    # 2.5937..., 2.7182... -> ~5% low, first order
```

RK4 is the same loop with the four-sample update in place of the single step:

```
def rk4(f, t0, y0, h, n):
    t = np.empty(n + 1); y = np.empty(n + 1)
    t[0], y[0] = t0, y0
    for k in range(n):
        k1 = f(t[k], y[k])
        k2 = f(t[k] + h/2, y[k] + h/2 * k1)
```

```

    k3 = f(t[k] + h/2, y[k] + h/2 * k2)
    k4 = f(t[k] + h, y[k] + h * k3)
    y[k + 1] = y[k] + h/6 * (k1 + 2*k2 + 2*k3 + k4)
    t[k + 1] = t[k] + h
return t, y

```

With the same $h = 0.1$, RK4's error at $t = 1$ is around 10^{-6} versus Euler's few percent — the fourth-order payoff in one experiment.

7.2 SciPy's `solve_ivp`

For real work, hand-written loops give way to `scipy.integrate.solve_ivp`, which implements adaptive RK (and stiff) methods with automatic step control. It also solves **systems**: pass a vector right-hand side and a vector initial condition.

```

from scipy.integrate import solve_ivp
import numpy as np

# A 2-D system: predator-prey-like linear dynamics y' = [[0,1],[-1,0]] y
def rhs(t, y):
    return [y[1], -y[0]]          # equivalent to y'' + y = 0

sol = solve_ivp(rhs, t_span=(0, 10), y0=[1, 0],
                method="RK45", t_eval=np.linspace(0, 10, 200))

# sol.t : times, sol.y : array of shape (2, 200) with the trajectories
import matplotlib.pyplot as plt
plt.plot(sol.t, sol.y[0])         # plots cos(t), the first component

```

Comparing methods. `solve_ivp`'s `method` argument selects the scheme: "RK45" (adaptive 4th/5th-order Runge–Kutta, the default), "RK23" (cheaper, lower order), "DOP853" (very high order), or "Radau"/"BDF" for **stiff** problems where explicit methods need impractically tiny steps. Solving the same problem with two methods and comparing trajectories — or checking against an analytical solution where one exists — is the standard way to gain confidence in a numerical answer.

8 Applications: finance and growth

The whole apparatus exists to model real dynamics. We close with the canonical finance and growth models named in the program — each an ODE, most solvable by the methods above, all validatable numerically.

8.1 Continuous compound interest

Interest compounded ever more frequently approaches a **continuous** limit. If a balance y earns annual rate r compounded continuously, in time dt it earns $ry\,dt$, giving the simplest growth ODE.

Worked example — The exponential of compounding.

$$y' = ry, \quad y(0) = y_0 \implies y(t) = y_0 e^{rt}.$$

Separable (or first-order linear): the balance grows exponentially. At $r = 5\%$, an investment doubles when $e^{0.05t} = 2$, i.e. $t = \ln 2 / 0.05 \approx 13.9$ years — the continuous analogue of the “rule of 72”.

8.2 Portfolio with contributions

Add a steady stream of contributions at rate c (currency per year) to the compounding balance. The rate of change is now interest **plus** deposits — a first-order linear, non-homogeneous equation.

Worked example — Saving while compounding.

$$y' = ry + c, \quad y(0) = y_0.$$

Integrating factor $\mu = e^{-rt}$ (or variation of constants) gives

$$y(t) = \left(y_0 + \frac{c}{r}\right)e^{rt} - \frac{c}{r}.$$

The term $-c/r$ is the equilibrium the deposits alone would sustain against no growth; the bracket carries the compounding of both the initial sum and the contribution stream. Setting $c = 0$ recovers pure compounding.

8.3 Logistic growth

Exponential growth is unrealistic in the long run — resources are finite. The **logistic** model adds a braking term that vanishes at a **carrying capacity** K : growth is near-exponential when y is small and stalls as $y \rightarrow K$.

Definition 22 The **logistic equation** with intrinsic rate r and carrying capacity K is

$$y' = ry\left(1 - \frac{y}{K}\right).$$

Its equilibria are $y = 0$ (unstable) and $y = K$ (stable).

Worked example — Solving the logistic equation. Separating variables and integrating (a partial-fraction step on $1/(y(1 - y/K))$) yields the **logistic curve**

$$y(t) = \frac{K}{1 + \left(\frac{K - y_0}{y_0}\right)e^{-rt}}.$$

As $t \rightarrow \infty$, $y \rightarrow K$ from below; the graph is the familiar S-shape, rising fastest as it crosses $y = K/2$.

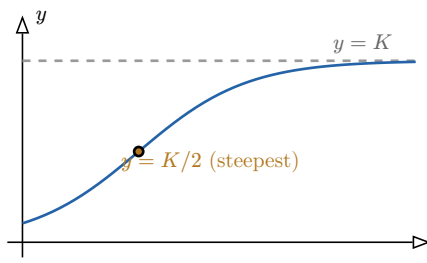


Figure 6: The logistic curve: near-exponential growth while y is small, an inflection (steepest growth) at $y = K/2$, then saturation toward the carrying capacity K (dashed). Contrast with unbounded exponential growth, which never levels off.

8.4 Debt repayment

A loan of balance y accrues interest at rate r while being repaid at a constant rate p (payment per year). The balance falls when payments outpace interest — structurally the same linear equation as the portfolio, with a sign flip.

Worked example — Paying down a loan.

$$y' = ry - p, \quad y(0) = y_0.$$

Solution $y(t) = (y_0 - p/r)e^{rt} + p/r$. The loan is cleared ($y = 0$) in finite time only if $p > ry_0$ — payments must exceed the initial interest ry_0 , or the balance grows without bound. Solving $y(T) = 0$ gives the payoff time $T = \frac{1}{r} \ln\left(\frac{p}{p - ry_0}\right)$.

9 Choosing a method: analytical or numerical?

Faced with a new ODE, the first decision is **how** to solve it. A misjudged choice wastes effort chasing a formula that does not exist, or reaches for the computer where a clean exact answer was one substitution away.

A decision checklist.

1. **Standard form & well-posed?** Write $y' = f(t, y)$; check that f and $\partial f / \partial y$ are continuous near the start, so a unique solution exists (Cauchy–Lipschitz) and is worth seeking.
2. **Recognise the type.** Is it separable? First-order linear? Second-order linear with constant coefficients? If yes, an **analytical** method applies — prefer it: an exact formula reveals structure (equilibria, growth rates, parameter dependence) no numerical table shows.
3. **Tractable in closed form?** Even a separable equation can lead to an integral with no elementary antiderivative (e.g. $y' = e^{-t^2}$). If the algebra stalls, switch to **numerical**.
4. **Otherwise, go numerical.** Use RK4 / `solve_ivp`; pick the step size by step-doubling, and validate against any analytical special case or a second method.

Worked example — Three equations, three verdicts.

- $y' = -2y + t$: first-order linear $\rightarrow$ **analytical** (integrating factor).
- $y' = y(1 - y)$: separable with a clean partial-fraction integral $\rightarrow$ **analytical** (logistic formula).
- $y' = \sin(ty) + t^2$: neither separable nor linear, no closed form $\rightarrow$ **numerical** (`solve_ivp`).

The honest default. The solvable catalogue of this course is small on purpose: it covers the equations that recur in finance and elementary growth models. The vast majority of ODEs that arise in practice are **not** in it, which is precisely why the numerical half matters as much as the analytical half — and why this course pairs them. When stochastic calculus later adds a random term to these dynamics, both skills carry over: the analytical intuition for structure, and the numerical machinery for everything structure cannot reach.